

20CYS403 – Web Application Security

XML, XML SCHEMA, JSON, AJAX AND FETCH API

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PART A – XML

Create an XML document containing the details of five students.

students.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<students>
  <student>
    <registerNumber>23034</registerNumber>
    <name>vedica</name>
    <department>CYS</department>
    <semester>6</semester>
    <cgpa>9.02</cgpa>
  </student>

  <student>
    <registerNumber>23045</registerNumber>
    <name>soffia</name>
    <department>CYS</department>
    <semester>6</semester>
    <cgpa>8.72</cgpa>
  </student>

  <student>
    <registerNumber>23052</registerNumber>
    <name>vedha</name>
    <department>CYS</department>
    <semester>6</semester>
    <cgpa>8.75</cgpa>
  </student>

  <student>
    <registerNumber>23002</registerNumber>
    <name>anagha</name>
    <department>CYS</department>
    <semester>6</semester>
    <cgpa>9.16</cgpa>
  </student>

  <student>
```

```

    <registerNumber>23334</registerNumber>
    <name>renny</name>
    <department>AI</department>
    <semester>6</semester>
    <cgpa>9.57</cgpa>
  </student>
</students>

```

PART B – XML Schema (XSD)

Modify the schema to ensure:

- Semester is between 1 and 8.
- CGPA is between 0 and 10.

Validation is successful here.

XML data to validate

```

23 <department>CVS</department>
24 <semester>6</semester>
25 <cgpa>8.75</cgpa>
26 </student>
27
28 <student>
29 <registerNumber>23002</registerNumber>
30 <name>anagha</name>
31 <department>CVS</department>
32 <semester>6</semester>
33 <cgpa>9.16</cgpa>
34 </student>

```

XML schema (XSD) data

```

14 <xsd:element name="name" type="xsd:string"/>
15 <xsd:element name="department" type="xsd:string"/>
16
17 <xsd:element name="semester">
18   <xsd:simpleType>
19     <xsd:restriction base="xsd:integer">
20       <xsd:minInclusive value="1"/>
21       <xsd:maxInclusive value="8"/>
22     </xsd:restriction>
23   </xsd:simpleType>
24 </xsd:element>
25
26 <xsd:element name="cgpa">

```

☒ Data Security Warning - Tick this box to confirm you understand all data is stored on our servers for processing and may be retained for service improvement.
Validate

If you have sensitive data, please [Download the Free Liquid Studio Community Edition](#) and use the tools directly from your desktop.

Document Valid

Since, we changed the cgpa to 45 which doesn't meet the schema requirement of it being in range of 0 to 10.

XML data to validate

```
23 <name>anagha</name>
24 <department>CYS</department>
25 <semester>4</semester>
26 <cgpa>8.75</cgpa>
27 </student>
28 <student>
29 <registerNumber>23882</registerNumber>
30 <name>anagha</name>
31 <department>CYS</department>
32 <semester>4</semester>
33 <cgpa>45</cgpa>
34 </student>
```

XML schema (XSD) data

```
14 <xsd:element name="name" type="xsd:string"/>
15 <xsd:element name="department" type="xsd:string"/>
16
17 <xsd:element name="semester">
18 <xsd:simpleType>
19 <xsd:restriction base="xsd:integer">
20 <xsd:minInclusive value="1"/>
21 <xsd:maxInclusive value="8"/>
22 </xsd:restriction>
23 </xsd:simpleType>
24 </xsd:element>
25
26 <xsd:element name="cgpa">
```

☒ Data Security Warning - Tick this box to confirm you understand all data is stored on our servers for processing and may be retained for service improvement.

Validate

Errors

Severity	Location	Filename	Message
	33:23	sample.xml	cvc-maxInclusive-valid: Value '45' is not facet-valid with respect to maxInclusive '10.0' for type '#AnonType_cgpastudentstudents'.

students.xsd

```
<?xml version="1.0" encoding="UTF-8"?>

<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">

  <xs:element name="students">
    <xs:complexType>
      <xs:sequence>

        <xs:element name="student" maxOccurs="unbounded">
          <xs:complexType>
            <xs:sequence>

              <xs:element name="registerNumber"
type="xs:string"/>

              <xs:element name="name" type="xs:string"/>
              <xs:element name="department"
type="xs:string"/>

              <xs:element name="semester">
                <xs:simpleType>
                  <xs:restriction base="xsd:integer">
                    <xs:minInclusive value="1"/>
                    <xs:maxInclusive value="8"/>
                  </xs:restriction>
                </xs:simpleType>
              </xs:element>
```

```

-                                     <xs:element name="cgpa">
-                                         <xs:simpleType>
-                                             <xs:restriction base="xs:decimal">
-                                                 <xs:minInclusive value="0"/>
-                                                 <xs:maxInclusive value="10"/>
-                                             </xs:restriction>
-                                         </xs:simpleType>
-                                     </xs:element>
-
-                                     </xs:sequence>
-                                 </xs:complexType>
-                            </xs:element>
-
-                        </xs:sequence>
-                    </xs:complexType>
-                </xs:element>
-
-            </xs:schema>

```

PART C – JSON

Create a JSON array containing five students.

students.json

```

[
  {
    "registerNumber": "23034",
    "name": "Vedica",
    "department": "CYS",
    "semester": 6,
    "cgpa": 9.02
  },
  {
    "registerNumber": "23045",
    "name": "Soffia",
    "department": "CYS",
    "semester": 6,
    "cgpa": 8.72
  },
  {
    "registerNumber": "23052",
    "name": "Vedha",
    "department": "CYS",
    "semester": 6,
    "cgpa": 8.75
  },
  {
    "registerNumber": "23002",
    "name": "Anagha",

```

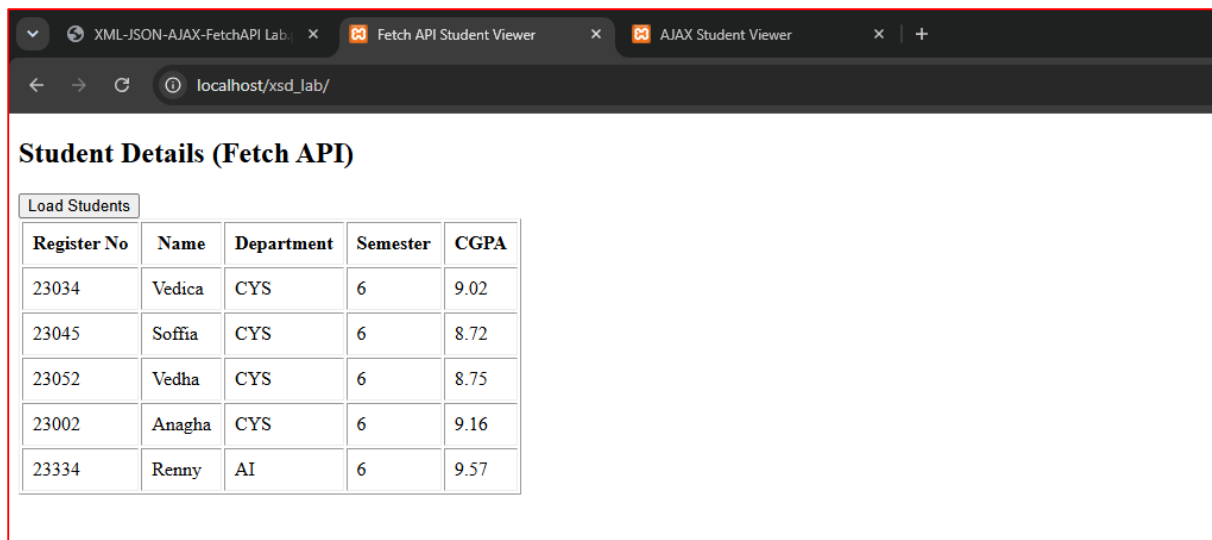
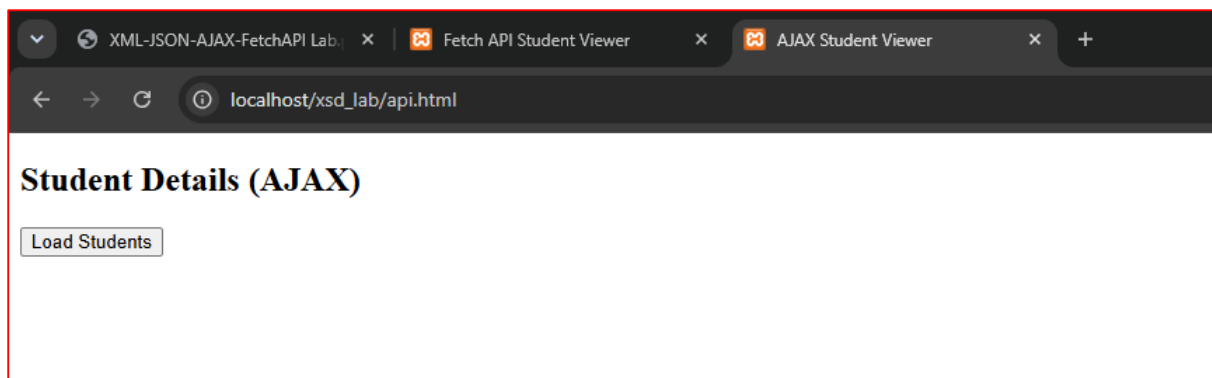
```

        "department": "CYS",
        "semester": 6,
        "cgpa": 9.16
    },
    {
        "registerNumber": "23334",
        "name": "Renny",
        "department": "AI",
        "semester": 6,
        "cgpa": 9.57
    }
]

```

PART D – AJAX

Modify the PHP script to return an array of students and display them in an HTML table.



The screenshot shows a web browser at `localhost:5051/api.html`. The page title is "Student Details (AJAX)". It features a "Load Students" button and a table with student data. The table has columns: Register No, Name, Department, Semester, and CGPA. The data is as follows:

Register No	Name	Department	Semester	CGPA
23034	Vedica	CYS	6	9.02
23045	Soffia	CYS	6	8.72
23052	Vedha	CYS	6	8.75
23002	Anagha	CYS	6	9.16
23334	Renuy	AI	6	9.57

The browser's developer tools are open to the Network tab, showing a request to `students.php` with a status of 200 and a type of `xhr`. The response is an XML document of size 0.7 kB, taking 4 ms to load.

Can see the network response as the *xhr* here

api.html

```
<!DOCTYPE html>
<html>
<head>
  <title>AJAX Student Viewer</title>
</head>
<body>

<h2>Student Details (AJAX)</h2>

<button onclick="loadStudents()">Load Students</button>

<div id="output"></div>

<script>

function loadStudents(){

  let xhr = new XMLHttpRequest();

  xhr.open("GET", "students.php", true);

  xhr.onload = function(){

    if(xhr.status == 200){

      let students = JSON.parse(xhr.responseText);
```

```

        let table="<table border='1' cellpadding='8'>";
        table+="<tr>";
        table+="<th>Register No</th>";
        table+="<th>Name</th>";
        table+="<th>Department</th>";
        table+="<th>Semester</th>";
        table+="<th>CGPA</th>";
        table+="</tr>";

        students.forEach(function(student){

            table+="<tr>";
            table+="<td>"+student.registerNumber+"</td>";
            table+="<td>"+student.name+"</td>";
            table+="<td>"+student.department+"</td>";
            table+="<td>"+student.semester+"</td>";
            table+="<td>"+student.cgpa+"</td>";
            table+="</tr>";

        });

        table+="</table>";

        document.getElementById("output").innerHTML=table;
    }
    else{
        document.getElementById("output").innerHTML="Unable to retrieve
student data.";
    }

    };
    xhr.send();
}

</script>

</body>
</html>

```

students.php

```

<?php

header("Content-Type: application/json");

$students = [

[

```

```
"registerNumber"=>"23034",
"name"=>"Vedica",
"department"=>"CYS",
"semester"=>6,
"cgpa"=>9.02
],

[
"registerNumber"=>"23045",
"name"=>"Soffia",
"department"=>"CYS",
"semester"=>6,
"cgpa"=>8.72
],

[
"registerNumber"=>"23052",
"name"=>"Vedha",
"department"=>"CYS",
"semester"=>6,
"cgpa"=>8.75
],

[
"registerNumber"=>"23002",
"name"=>"Anagha",
"department"=>"CYS",
"semester"=>6,
"cgpa"=>9.16
],

[
"registerNumber"=>"23334",
"name"=>"Renny",
"department"=>"AI",
"semester"=>6,
"cgpa"=>9.57
]

];

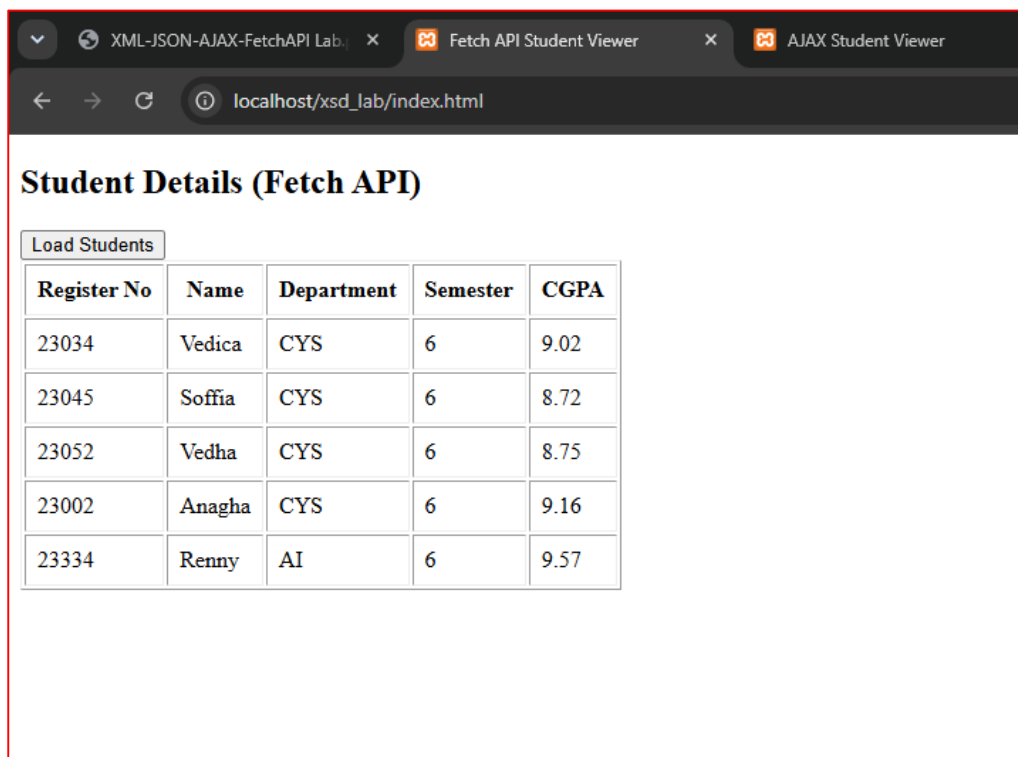
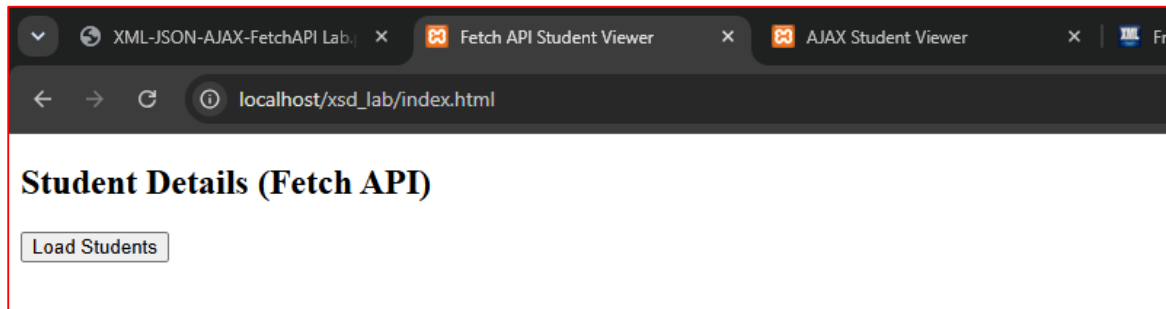
echo json_encode($students);

?>
```

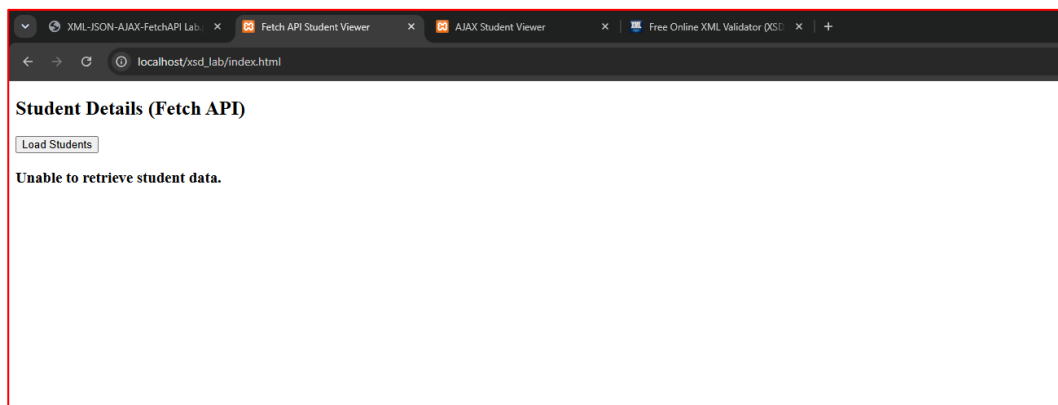
PART E – Fetch API

Modify the application so that:

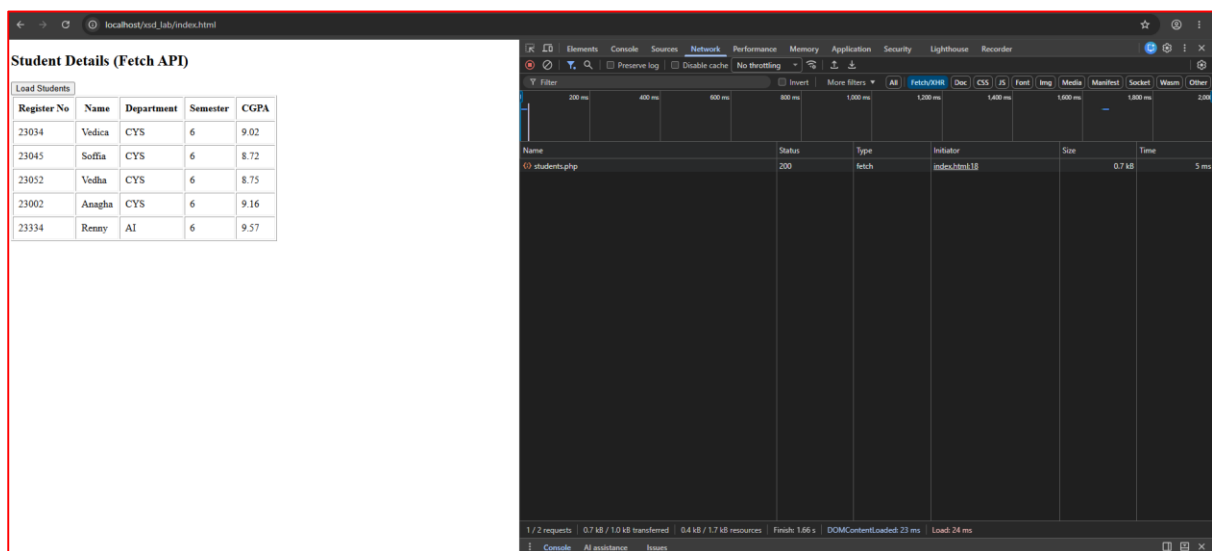
1. The PHP script returns details of five students.
2. The Fetch API retrieves the data.
3. The student information is displayed in an HTML table.
4. Display an appropriate error message if the server cannot be reached.



If server can't be reached, an error message is showed



Here, we can see that the *fetch* in network response



index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Fetch API Student Viewer</title>
</head>
<body>

<h2>Student Details (Fetch API)</h2>

<button onclick="loadStudents()">Load Students</button>

<div id="output"></div>

<script>
```

```
function loadStudents(){
  fetch("students.php")
    .then(response=>{
      if(!response.ok){
        throw new Error("Server Error");
      }

      return response.json();
    })

    .then(data=>{

      let table=<table border='1' cellpadding='8'>;

      table+=<tr>;
      table+=<th>Register No</th>;
      table+=<th>Name</th>;
      table+=<th>Department</th>;
      table+=<th>Semester</th>;
      table+=<th>CGPA</th>;
      table+=</tr>;

      data.forEach(student=>{

        table+=<tr>;
        table+=<td>+student.registerNumber+</td>;
        table+=<td>+student.name+</td>;
        table+=<td>+student.department+</td>;
        table+=<td>+student.semester+</td>;
        table+=<td>+student.cgpa+</td>;
        table+=</tr>;

      });

      table+=</table>;

      document.getElementById("output").innerHTML=table;
    })

    .catch(error=>{

      document.getElementById("output").innerHTML=<h3>Unable to retrieve
      student data.</h3>;
    })
  }
}
```

```
    console.log(error);  
  });  
}  
  
</script>  
  
</body>  
</html>
```